

Prime-Indexed Return Sectors

The Thirty-Six-Relation Closure of a Nine-State Carrier

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Abstract

A finite observer-closure theorem is developed from an occurrence-sensitive prime relation. The two appearances of a prime in $p \cdot 1 = p$ have equal numerical value but different source and return roles. A value-only reporter turns their directed relation into a self-loop; an occurrence-sensitive reporter retains a period-two return. On the decimal terminal carrier $\mathbb{Z}_{10}$, the zero and unit are terminal roles, while the eight nonterminal indices form four prime/composite pairs under the fixed-point-free affine involution $\sigma(d) = 1 - d \pmod{10}$. Removing the zero terminal leaves nine active states. Their complete unordered relation frontier contains $\binom{9}{2} = 36$ cells and decomposes exactly into eight unit relations, four return-sector relations, and twenty-four transverse relations. The four prime labels 2, 3, 5, 7 are independently selected by transparent arity. Together with a side label they form complete commuting spectral coordinates on the finite carrier; the four return planes commute inside the 36-dimensional antisymmetric relation space. The finite prime/composite reporter is then refined by prime-occupation coordinates: 6 is squarefree coupling, whereas 4, 8, 9 are repeated-prime states. The resulting static five-class reporter is proved insufficient under arbitrary prime multiplication, and the all-scale predictive quotient is the finite prime-support address plus one absorbing repeated-prime state. The theorem is related to the earlier nine-transformation, thirty-six-witness closure, recurrence selection, quotient closure, harmonic realization, and arithmetic completion without identifying merely isomorphic carriers as the same physical system.

Keywords: prime arity; predictive quotient; occurrence identity; relation lattice; recurrence exponent; finite spectral labels

Scope and status: The carrier, involution, 36-cell decomposition, reporter-closure, critical-return, occupation-state, and finite predictive-quotient results are exact. The term quantum number is used first in its mathematical spectral-label sense. A physical realization requires independently specified observables and dynamics.

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1 Occurrence-sensitive prime relation

For a prime p , ordinary arithmetic writes

$$p \cdot 1 = p. \quad (1)$$

The two tokens labelled p have one value but two roles. Write

$$p_{\text{source}} \cdot 1_{\text{unit}} = p_{\text{return}}, \quad v(p_{\text{source}}) = v(p_{\text{return}}) = p, \quad p_{\text{source}} \neq_{\text{occ}} p_{\text{return}}. \quad (2)$$

The fourth component is the relation-establishing operation and its retained mark. It is not another numerical factor.

Proposition 1.1 (Factor-complete prime triad). *For an integer $n > 1$, the reduced triad $(n_{\text{source}}, 1_{\text{unit}}, n_{\text{return}})$ contains every positive factor relation of n if and only if n is prime.*

Proof. If n is prime, every positive factor pair is $(1, n)$ or $(n, 1)$. If n is composite, some $1 < a, b < n$ satisfy $ab = n$, and that relation is absent from the reduced triad. $\square$

The proposition is deliberately modest. Its significance lies in reporter design: a terminal primality report is complete for the declared positive-factor query, while a generative prime-power or multiplication query requires occurrence and multiplicity coordinates.

2 Terminal lattice and affine return

Let

$$D = \mathbb{Z}_{10}, \quad T = \{0, 1\}, \quad I = \{2, 3, 4, 5, 6, 7, 8, 9\}, \quad A = D \setminus \{0\}. \quad (3)$$

Define

$$\sigma(d) = 1 - d \pmod{10}. \quad (4)$$

It has no fixed point on D and has the five orbits

$$(0, 1), (2, 9), (3, 8), (4, 7), (5, 6). \quad (5)$$

The restriction to I pairs every prime index with one composite index. Put

$$P = \{2, 3, 5, 7\}, \quad C = \{4, 6, 8, 9\}. \quad (6)$$

For $q \in P$, define $\bar{q} = \sigma(q)$ and encode the interior as

$$I \cong P \times \{+, -\}, \quad (q, +) = q, \quad (q, -) = \bar{q}. \quad (7)$$

Then

$$\sigma(q, \varepsilon) = (q, -\varepsilon). \quad (8)$$

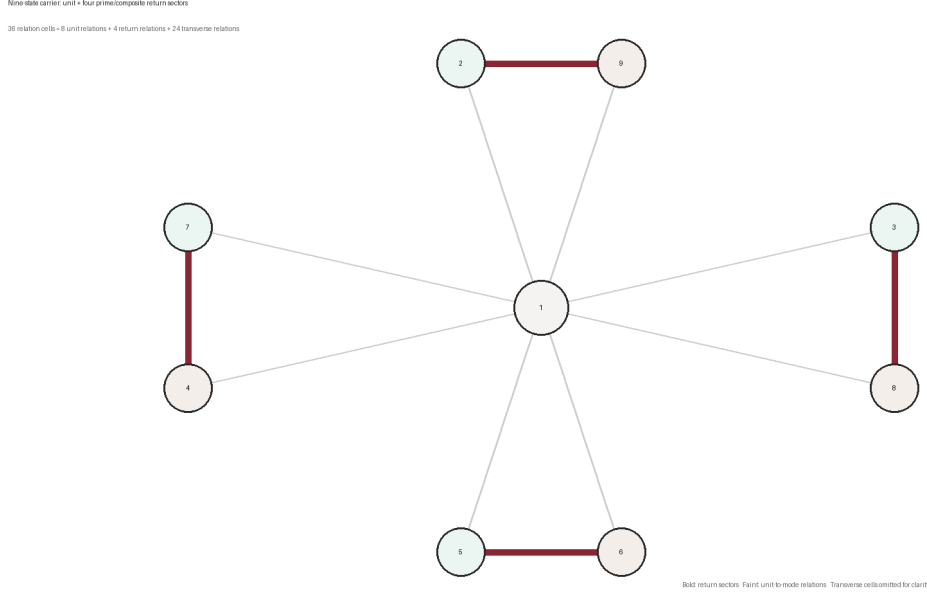


Figure 1: The active nine-state carrier. Bold edges are the four prime/composite return sectors. The eight unit relations are shown faintly; the twenty-four transverse relation cells complete the rank-two frontier.

3 Thirty-six-relation closure theorem

Let

$$\mathcal{R}_2(A) = \{\{x, y\} : x, y \in A, x \neq y\} \quad (9)$$

be the complete unordered pair-relation frontier.

Theorem 3.1 (Nine-state, thirty-six-relation closure). *The active carrier has nine states and*

$$|\mathcal{R}_2(A)| = \binom{9}{2} = 36. \quad (10)$$

The frontier decomposes canonically as

$$36 = 8_{\text{unit}} + 4_{\text{return}} + 24_{\text{transverse}}. \quad (11)$$

The four return relations are $\{q, \bar{q}\}$ for $q \in P$. The transverse relations connect different return sectors.

Proof. There are eight relations from the unit state 1 to the eight interior states. There are four within-sector relations. For each of the $\binom{4}{2} = 6$ pairs of two-element sectors, there are $2 \cdot 2 = 4$ cross-sector relations, giving 24. The classes are disjoint and exhaustive. $\square$

This is more accurately a 36-cell rank-two relation layer than a 36-state carrier. Each cell is one possible distinction between two of the nine active states.

4 Observer quotient and critical recurrence

Define the side-forgetting reporter

$$U(1) = 1, \quad U(q, \varepsilon) = q. \quad (12)$$

Let the return map be

$$R(1) = 1, \quad R(q, \varepsilon) = (q, -\varepsilon). \quad (13)$$

Then $U \circ R = U$, although R has period two on every interior sector.

Theorem 4.1 (Apparent halt versus retained return). *For every $q \in P$, the complete state alternates*

$$(q, +) \leftrightarrow (q, -), \quad (14)$$

while the quotient report remains constant. If $r_q = e_{q,+} - e_{q,-}$ and the residual propagator is the return flip, then

$$Rr_q = -r_q, \quad \|R^n r_q\| = \|r_q\|, \quad (15)$$

and therefore $\chi(r_q) = 0$ for every linear intrinsic clock.

Proof. The first statement is immediate. The residual changes sign under the swap and therefore has constant norm. Its logarithmic growth rate is zero. $\square$

Thus a reporter that stops when its visible output repeats records a halt; a reporter that preserves occurrence sees recurrence. The equality $U(r_q) = 0$ is report-level erasure, not dynamical assimilation. If an admitted future distinguishes the two sides, the complete-behaviour closure criterion forces the refined state (q, ε) .

5 Prime selection and finite spectral labels

Transparent arity is defined by

$$\binom{q}{k} \equiv 0 \pmod{q} \quad (0 < k < q). \quad (16)$$

The transparent arities are exactly the primes. Hence 2, 3, 5, 7 are not selected by the decimal drawing; they are the first four transparent nonterminal arities in the finite carrier.

Let

$$\mathcal{H}_A = \mathbb{C}|1, 0\rangle \oplus \bigoplus_{q \in P, \varepsilon = \pm 1} \mathbb{C}|q, \varepsilon\rangle. \quad (17)$$

Define commuting diagonal operators

$$\widehat{Q}|q, \varepsilon\rangle = q|q, \varepsilon\rangle, \quad \widehat{S}|q, \varepsilon\rangle = \varepsilon|q, \varepsilon\rangle. \quad (18)$$

The pair (Q, S) is a complete finite label. In this exact mathematical sense it is a system of quantum numbers: joint eigenvalues of commuting operators. Physical quantum numbers require a specified physical observable and transition law; the finite labelling theorem alone does not provide them.

The full oriented pair-relation space is

$$\bigwedge^2 \mathbb{R}^9 \cong \mathfrak{so}(9), \quad \dim \bigwedge^2 \mathbb{R}^9 = 36. \quad (19)$$

For each return sector define

$$J_q = E_{q, \bar{q}} - E_{\bar{q}, q}. \quad (20)$$

The four J_q act on disjoint two-planes and therefore commute. This is a finite relation representation, not a derived physical $SO(9)$ gauge theory.

6 Prime-occupation refinement

Unique factorisation gives commuting occupation labels

$$N_p|d\rangle = v_p(d)|d\rangle, \quad p \in \{2, 3, 5, 7\}. \quad (21)$$

The finite states are therefore separated as follows.

index	(n_2, n_3, n_5, n_7)	type
1	(0, 0, 0, 0)	unit
2, 3, 5, 7	one unit occupation	primitive prime mode
6	(1, 1, 0, 0)	squarefree coupling
4, 8, 9	a repeated occupation	repeated-prime state

Accordingly the coarsest terminal-resolved static Möbius reporter on the digit carrier has five classes

$$\{0\}, \quad \{1\}, \quad \{2, 3, 5, 7\}, \quad \{6\}, \quad \{4, 8, 9\}. \quad (22)$$

Theorem 6.1 (Static five-class false closure). *The five-class reporter is not a continuation congruence when multiplication by arbitrary primes is admitted. The coarsest all-prime predictive quotient for future Möbius reports consists of one state for every finite squarefree prime-support set and one absorbing repeated-prime state.*

Proof. The primes 2 and 3 share the static prime class. Multiplication by 2 sends them to 4 and 6, whose Möbius reports are 0 and +1. Hence the static quotient fails. For the full quotient, represent a squarefree integer by its finite prime support and every nonsquarefree integer by $\dagger$. Multiplication by p adjoins p when absent and sends the state to $\dagger$ when present; $\dagger$ is absorbing. Distinct supports are future-distinguishable: choose a prime in their symmetric difference and multiply by it. One state repeats that prime and the other does not. Therefore no coarser reporter preserves every prime-multiplication future. $\square$

This theorem locates the limit of the finite 36-cell picture. It is a complete finite relation frontier, but the all-prime predictive state is necessarily unbounded in the number of possible support addresses.

7 Harmonic realization and scale boundary

For any positive fundamental f_0 , define

$$f_d = f_0 d, \quad d = 0, 1, \dots, 9. \quad (23)$$

At $f_0 = 111$ Hz the coefficients appear as decimal repdigits. The prime-indexed and composite-indexed modes are

$$111\{2, 3, 5, 7\} \quad \text{and} \quad 111\{4, 6, 8, 9\}. \quad (24)$$

Their indicator sequences are affine reflections and have identical cyclic Fourier magnitudes. Equal-amplitude multitone realizations are equal-energy and orthogonal over the common period $1/f_0$. The structure is invariant under positive rescaling; therefore it does not select 111 Hz as a natural constant, nor does the arithmetic alone imply a biological or therapeutic effect.

8 Relation to the earlier nine-transformation closure

A separate finite construction generated nine causally distinguishable whole-carrier transformations and an exact pairwise witness for each of their 36 unordered pairs. Both systems therefore have a complete relation frontier isomorphic to the edge set of K_9 .

Proposition 8.1 (Relation-frontier isomorphism and its limit). *Any bijection between two nine-element carriers induces an isomorphism of their 36-cell unordered relation frontiers. It preserves operational meaning only when it also intertwines the declared continuations and source provenance.*

Proof. A bijection maps unordered pairs bijectively. Graph isomorphism alone says nothing about continuation responses. Those are preserved exactly when the vertex map commutes with the relevant operations and reporters. $\square$

Thus the two nine-state constructions reinforce the role of complete pairwise distinction, but combinatorial isomorphism is not used as evidence that their vertices or physical meanings are identical.

9 Integration with the wider programme

The theorem joins the earlier work in one precise chain:

$$\text{occurrence} \rightarrow \text{relation} \rightarrow \text{report} \rightarrow \text{future test} \rightarrow \text{assimilation, retention, or promotion.} \quad (25)$$

The prime triad supplies the smallest occurrence-sensitive example. The 36-cell frontier enumerates every pair distinction on the nine active states. The recurrence exponent distinguishes genuine assimilation from report-level erasure. Prime transparency selects the sector labels. Occupation coordinates refine the composite class. The all-prime quotient shows that a finite reporter can be exact for one envelope and false for a larger one. The adaptive-system architecture turns this sequence into an executable repair protocol.

10 Verification and claim boundary

The ancillary verifier checks the affine involution, the four prime/composite pairs, the $8 + 4 + 24$ relation decomposition, commutation of the four return-plane generators, Fourier-magnitude equality, the static five-class counterexample, and deterministic prime-support updates.

The paper does not claim that nature has exactly nine states, that $SO(9)$ is a physical gauge group, that 2, 3, 5, 7 are measured physical quantum numbers, that 111 Hz is fundamental, or that the finite carrier proves the Riemann Hypothesis. It establishes the finite theorem, the exact predictive-closure correction, and the conditions required for a physical realization.

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